

Optimization of a UAV for Wildfire Management

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Dissertation submitted to the Faculty of the
Virginia Polytechnic Institute and State University
in partial fulfillment of the requirements for the degree of

Master of Science
in
Mechanical Engineering

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December XYZ, 2025

Blacksburg, Virginia

Keywords: Some Keywords, Subject matter, etc.

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(ABSTRACT)

Give a brief description of your thesis here.

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(GENERAL AUDIENCE ABSTRACT)

You are also required as of Spring 2016 to include a general audience abstract. This should be geared towards individuals outside of your field that may be reading seeking information about your work. You should avoid language that is particular to your field and clearly define any terms that may have special meaning in your discipline.

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NLP is a field of computer science, artificial intelligence, and linguistics concerned with the interactions between computers and human (natural) languages.

σ is the eighteenth letter of the Greek alphabet, and carries the 's' sound. In the system of Greek numerals, it has a value of 200.

Chapter 1

Introduction

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1.1 One Section

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1.1.1 A sub-section

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Chapter 2

Review of Literature

Chapter 3

UAV Design

3.1 UAV Geometry

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3.2 Mission Profile

3.3 UAV Batteries

Chapter 4

Finite Element Analysis

In order for the optimization algorithm to run repeatedly, a finite element beam model was created. The finite element solver is based on a Timoshenko beam theory. This formulation accounts for transverse shear deformation, which is important when analyzing beams with a short cross-section.

4.1 UAV Segment Model

The FEA model is a single slice of the UAV with symmetric constraints at the midpoints of the strut beams. This allows for faster analysis, and ignores the asymmetric modes, which were analyzed separately. Each beam is defined as an annular cross section. The hub is represented as a series of repeatedly narrowing I-beam cross sections. Each hub beam is labeled

"hub_{beam_x}." The number of hub sections is arbitrary and is increased until the solution converges. Attached to the section of the beam. Finally, an annular arm beam spans from the point where the strut and arm meet to the outer

[include figure]

4.2 Beam Element Formulation

Each beam element is a Timoshenko beam. This beam model takes shear deformation effects into account. The mass for each beam is modeled as a 3D lumped mass matrix.

Each node is stored as a Node object. The Nodes

4.3 Solvers

4.3.1 Static

Forces are represented as point loads at the end of the UAV beam.

4.3.2 Dynamic

4.3.3 Transient

4.4 Optimization

The problem is formulated as follows:

$$\min \text{mass}(d_a, t_a, d_s, t_s, r_s, t_f, t_w)$$

subject to:

$$g_j(x) \leq 0; j = 0, 1, 2, \dots, J$$

$$x_i^L \leq x_i \leq x_i^U$$

design variables:

- d_a : arm diameter,
- t_a : arm thickness,
- d_s : strut diameter,
- t_s : strut thickness,
- r_s : strut distance from center,
- t_f : hub flange thickness,
- t_w : hub web thickness

derived variables:

- d_{tip} : Deflection at tip
- σ_{max} : Maximum stress from bending
- f_{amp} : Amplitude of displacement at the tip of the uav arm
- D : Damage per mission
- E_{used} : Energy used over th duration of the mission
- E_{total} : Energy used over th duration of the mission

design parameters:

- $d_{allowable}$: Deflection allowable
- $\sigma_{allowable}$: Stress allowable from bending
- $R_{allowable}$: Allowable amplitude displacement ratio
- $n_{missions}$: numbr of missions before damage

Constraints:

Limit on deflection from bending by force required to lift the UAV.

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = d_{tip} \leq d_{allowable}$$

Limit on stress from bending by force required to lift the UAV.

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \sigma_{max} \leq \sigma_{allowable}$$

Limit on amplitude of displacement given a forcing function based on the propeller speed of the UAV.

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Limit on the damage the UAV during a mission.

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = D * n_{missions} \leq 1$$

Energy

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = E_{used}/E_{total} \leq 0.8$$

4.4.1 Objective Function and Design Variables

4.4.2 Constraints

Displacement

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = d_{tip} \leq d_{allowable}$$

Stress

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \sigma_{max} \leq \sigma_{allowable}$$

Frequency

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Damage

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = D * n_{missions} \leq 1$$

Energy

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = E_{used}/E_{total} \leq 0.8$$

Chapter 5

Results

Chapter 6

Conclusions

Chapter 7

Summary

Appendices

Appendix A

First Appendix

A.1 Section one

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A.2 Section two

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Appendix B

Second Appendix

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